Multivariate Sales Forecast Model Towards Trend Shifting During COVID-19 Pandemic

A Case Study in Global Beauty Industry

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ABSTRACT

COVID-19 pandemic has changed the economic weather and business performance in multiple streams. The uncertainty condition caused by the pandemic needs to be carefully taken care by all companies and organizations due to rapid consumer trend shifting and volatile market condition. The sales and marketing strategy needs to be carefully taken during organizational decision-making process to avoid further loss. PT XYZ as one of the leading consumer goods in beauty industry experiences the same condition and challenge reflected by down-trend in the organization KPI. This research aims to introduce and provide predictive data analytics tools for enhancing sales forecast by comparing Random Forest and Neural Network as part of machine learning methods also Vector Autoregression (VAR) as conventional statistical forecasting methodology. As the result of this research, neural network returns better evaluation for skin care and Vector Autoregression for makeup category. Meanwhile data visualization is found necessary to provide additional factual information, includes the external factor, to support knowledge management for better rational decision-making process.

CCS CONCEPTS

• **Information systems** → Information systems applications; Decision support systems; Data analytics.

KEYWORDS

COVID19, Sales Forecast, Multivariate, Machine Learning, Data analytics, Decision Making

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1 BACKGROUND

Covid-19 is considered as a critical disruptive event that cause global effect on several areas, but mainly on society and economy [1]. For retail and consumer goods industry, several studies have indicated changes on consumer and transactional behavior [2] [3] [4]. Most of the research result indicate pandemic has changed consumer behavior and decision making on the product needs [3] and way of transactions from offline to online [5]. For retail industries, the product needs shifting has a significant effect on their business. Pet and food supplies, house hold goods, and cosmetic product are the product categories with sustainable growth during pandemic meanwhile in the other hand the out of home entertainment and travel are the product categories affected negatively [6].

During market uncertainty caused by COVID-19 pandemic, PT. XYZ as one of the leading beauty industries, are currently facing a challenge on its financial key performance indicator in several categories across its product assortment, mainly in makeup category. Based on the discussions and available descriptive analytics, the challenge on financial KPI is mainly related with the consumer trend shifting which causing inaccurate organization strategy on doing marketing investment and supply chain planning. The supply chain planning plays important role on determining the stock based on sales forecast number. Sales forecast number is typically generated using simple statistical methodology by using historical sales and inventory data obtained from ERP System.

By using PT XYZ as case study, this research aims to introduce other methods on forecasting sales volume by not only utilizing internal ERP data in order to enhance sales and marketing strategy (univariate), but also utilizing external factor or exogenous variable in the forecast. This research also explains on data analytics as source in knowledge creation data-information-knowledge process [7]. In order to express the knowledge management variable, expert user tacit knowledge is also being taken into account as variable, i.e. the sales seasonality insight, as well as the special promotion upscaling (if any). Based on the pre-research investigation and interview, this study will focus on sales forecast area in the organization. Besides that, this study will complement academically current limited literature on usage of data analytics to ERP data and its role as source of knowledge management and impact to sales forecast number as organizational performance.

This research is expected to answer below questions to align with research objectives:

- What is the best forecasting model for sales forecast tools based on the evaluation measurements?

- What are parameters or features to be displayed in data visualization that can be used to interpret and reflect the sales forecast accuracy improvement compared to actual sales for organization strategy?

2 LITERATURE REVIEW

This chapter introduces basic understanding and foundation of ERP, machine learning which part of data analytics and the relationship to knowledge management and decision making, sales forecast as one of the organizational performance indicators, and also a brief explanation on COVID-19 in Jakarta Raya Area. In addition to that, high level understanding on the case study, including the enterprises and theoretical view and literature of the case study scope will also be expressed in this chapter.

2.1 Enterprise Resource Planning Data

ERP is a very important information system tool for corporation in order to manage the business processes by identify, capture, integrate, and store the flow of information as result of executing business transactions [8]. With multiple modules spans from supply chain up to finance that ERP has, ERP provides the capabilities to control information flows [9]. Based on previous studies and experience, ERP is having positive impact on its usage to the organization performance by bringing competitive advantage, it ensures cost reduction, better resource controls, improvement on decision quality, and streamlined business process [10]. In the relationship with Knowledge Management, ERP companies provide products to the client to facilitate knowledge management in whole lifecycle: Choosing ERP, Implementing ERP, and post the implementation – using and utilizing ERP [11].

2.2 Data Analytics in the Organizations

Nowadays, data is something massively available in the organization. It is then influenced many organizations in various business stream adopt data analytics and machine learning into their firm. As the result of this popular trend, data science is introduced as new discipline that discuss on how to derive insight and knowledge from the data [12]. Many organizations are interested to apply data analytics further known as big data analytics due to its positive impact to the organizations. Big data analytics is known to enhance decision making process of many businesses with recent methodology and revolutionary approach [13]. Besides improving decision making, big data analytics also improve learning and innovation in which lead to better organizational performance [14].

2.2.1 Data Analytics, Knowledge Management, and Decision Making. Big data analytics is one of the sources of knowledge creation in the organization [15]. The activity of data analytics is fundamentally about extracting information and ultimately knowledge from the raw data, this model is actually known as data-information-knowledge-wisdom model [16]. Several previous studies were exploring the relationship and map between big data analytics and knowledge management. Based on [17] big data is actually a part on knowledge management, an element to be precise. As shown in Figure 1 below, the part of 'data- information-knowledge' in data analytics is also considered as knowledge life cycle in knowledge management.

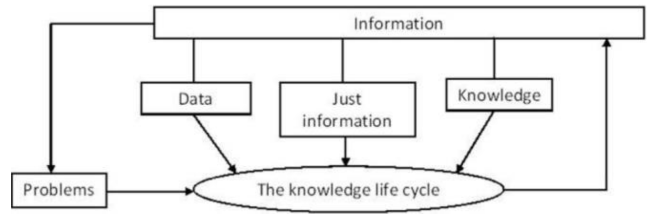


Figure 1: The Knowledge Lifecycle [17]

Wang and Wang in their research described the big data resource management is actually a form of knowledge management since big data act as major source of the explicit knowledge [18]. Another view on the relationship between big data analytics and KM is on how actually knowledge management contributes to big data analytics. The study explain that human knowledge develops the capabilities of big data analytics, by being responsible for decisions of selecting the data and analytics tools [19], [20]. Based on number of previous studies, it is indicated that big data and knowledge management are always correlated. Pauleen, in his research stated that big data can only generate big knowledge when Knowledge Management is involved [20], and enterprise can actually experience knowledge creation by using big data analytics [21].

There is a direct relationship between KM and decision-making [22]. This relationship explained as every step of decision making is influenced by knowledge management [23]. [24] in the research define two kinds of decision-making style in the organization, intuitive decision-making style and rational decision-making style. There are three elements as source of intuitive decision-making, 1) instinct or innate response, 2) general experience, and 3) focused learning [25]. In rational decision-making, comprehensive information collection is involved as well as analysis through factual information and evidence. In the organization when the rational decision-making style is involved, it is expected that business users are expected to evaluate both internal and external environment to make strategic decisions based on unbiased criteria and systematic analysis [24].

2.2.2 Predictive Data Analytics: Forecasting Methodology. Sales forecast in the organization is known as predictive analysis for future organizational strategy. This predictive analysis helps to predict future product sales by analyzing past historical data and current market data [26]. Accurate sales forecast may lead into accurate production planning in which will increase the company's revenue and also reducing cost. Prior to the trends of data analytics, many organizations or retail are using statistical- based model for univariate and multivariate time series forecast, such as linear regression, quantile regression, moving average (ARIMA, ARMA, SARIMA) [27], Vector Auto Regression, and many more [28]. But then after data analytics become popular, several machine learning techniques are introduced via many various studies [26]. Neural network techniques was applied by Kuo for forecasting [29], including recurrent neural network for sales forecast [30]. Forecasting for one of the electric companies in china are using clustering, regression, and time series analysis techniques [31]. Advanced Machine Learning techniques are also introduced such as SVR (support vector regression) for prediction of IT Tools [32].

2.3 Sales Forecast and COVID-19

Since the first case was found in China on December 2019, the covid-19 outbreak has infected more than 346 million people globally and bring deaths to over 5 million members of the world population [33]. One of the most expected outcomes as the impact of the pandemic is the economic situation all over the world, including Indonesia. Latest research conclude that the COVID-19 outbreak will reduce of Indonesia economic growth rate by 0,8% at its best, or even reduce 8.5% at its worst scenario. The research also estimates the declining of per-capita household expenditure due to slower economic growth and fewer economic activity [34].

The declining per-capita household expenditure due to COVID-19 as mentioned previously is obviously causing disruption in number of consumers sales for certain products. In the other hand, changes of consumer behavior of buying goods are also impacting on adjustment in market sales for certain products (for example healthcare and food product sales increase significantly, furthermore due to panic buying). It is important for many organizations now to understand and face the challenges of very dynamic consumer sales pattern during the pandemic [36] to have better sales and marketing strategy in order to still achieve competitive advantage.

As the COVID-19 cases number are unpredictable, several forecast methodologies have been taken into account as predictive tools. Some of them are sales excess [36] or customer satisfaction [1] The usage of the predictive tools may vary based on the needs and the context of the information. For instance, the predictive tools on forecasting the covid-19 cases will be suitable for the related government / health official taking and firming appropriate decisions [35]. Meanwhile the predictive tools on consumer sales might be beneficial for organization on deciding supply chain and commercial strategy in order to gain competitive advantage.

2.3.1 Multivariate Forecasting Methodology. There are several forecasting methodologies that consider several variables to build forecast number, known as multivariate forecast method. In multivariate forecasting methodology, the prediction model will be built not only based on endogenous variable but also exogenous variables. One of the popular multivariate time series forecasting is Vector Autoregression (VAR). VAR is a multivariate time series statistical model that consider multiple variables are dependent for building a prediction model of one target variable. In VAR, each variable uses its own past value and other variable past values in the system [37]. VAR model with k variables and p lags can be expressed mathematically as

$$Y_t = B_0 + BY_{t-1} + BY_{t-2} + \ldots + B_p Y_{t-p} + \epsilon_t \quad (1)$$

Vector Autoregression is a well-known statistical time series modelling for generating forecast number. But however, VAR has a pre-requisite of non-seasonality data, this means the variables being forecasted must be stationary, which can be tested using Augmented Dickey Fuller Test (ADF). A time series is considered stationary if its mean and standard deviation are stable and steady

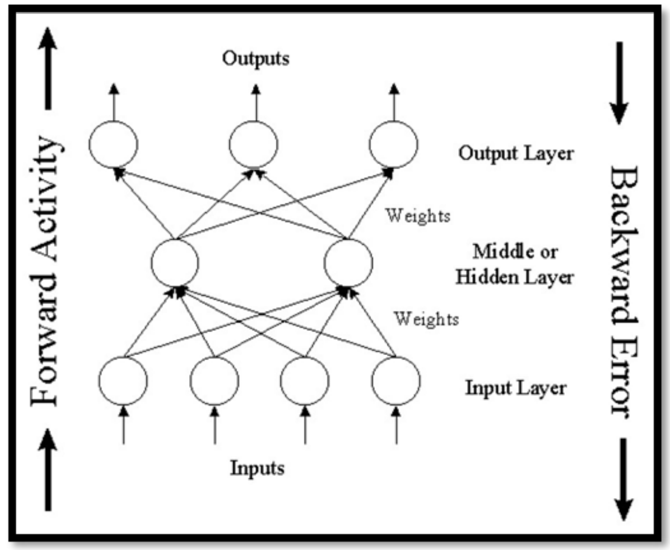


Figure 2: Neural Network Model [39]

over the time. If a time series is stationary then it has a larger possibility in repeating the behavior in the future, hence, will be easier to forecast. The most famous method for achieving stationary of time series is transforming the dataset by doing differencing. Besides statistical-based multivariate forecast, machine learning based (supervised learning) forecast is also being used to predict future value of target variable by using numbers of other variables (multivariate). One of the most popular machine learning methods is Artificial Neural Network (ANN). ANN is widely being used for sales forecast because of its flexible behavior on detecting data pattern [38].

As discussed previously ANN can learn from the multi-variable data and identify the pattern and predict future value. ANN model can be written mathematically as follow:

$$y_t = \alpha_0 + \sum_{j=1}^{n} \alpha_j f\left(\sum_{i=1}^{m} \beta_{ij} y_{t-i} + \beta_{0j}\right) + \varepsilon_t \quad (2)$$

With number of nodes in the input layer are denoted by m and in hidden layers. Figure 2 illustrate standard neural network model.

Other machine learning method that is known works for multivariate forecasting is random forest. Random Forest is a tree-based models and use splitting in order to build a model based on ensemble of univariate tree [40]. Even though random forest is a tree-based classifier, since it is ensemble of tree, the number of splits and calculation of impurity and happening across variables. The variables used to build each tree are also randomly selected from the total set of input variables available. [41]. The dataset is first split into two sets of data, train data and test data, there are several methods on doing this, but the simplest one is known as common split [27], [42] in which splitting the train and test data by percentage (75% and 25%, 80 and 20%, etc). The model then be tested by test set by using several evaluation criteria in which will be discussed in next section.

2.3.2 Forecast Model Evaluation. There are several measurements being used for evaluating forecast model. But the most common and popular for time series forecast is MSE, RMSE, and MAPE. These measurements is also being used and discussed in some research related to forecast [27], [43]. Mean Square Error is defined as the variance between predicted and actual values (average). The lower MSE values the better model is. If the MSE value is 0.00, it means no error value occurs in the prediction model.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(P_i - A_i)^2 \tag{3}$$

Where n is the number of samples, P is the predicted value and A is the actual value. Root-Mean Square Error represents the error-index, which is used to compare different forecasting models. This is the square root of the square differences measured between prediction and actual observation.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(P_i - A_i)^2} \tag{4}$$

Where n is the number of samples, P is the predicted value and A is the actual value. MAPE (Mean Absolute Percentage Error) is one of the popular and preferred evaluation criteria of sales forecast. The opposite of MAPE, is widely known as forecast accuracy.

$$MAPE = \frac{100\%}{n}\sum_{t=1}^{n}\left|\frac{A_t - F_t}{A_t}\right| \tag{5}$$

where At is the actual value and Ft is the forecast value.

3 EXPLORATORY DATA ANALYSIS

Prior to build the forecasting tools, understanding each data and features is performed through exploratory data analysis. The transformed data set contains 288 data points from originally 1288 data points. In the initial data we have 11 features represented as data column, they are row labels, positivity percentage, EUR to IDR, SKIN STOCK, SKIN SI, MAKE SI, MAKE STOCK, SKIN ACTUAL SO, MAKE ACTUAL SO, SOCIAL, SPECIAL, and FESTIVE. The data analysis is initiated by plotting the target variables in which the sales out data for both skin care and makeup product category to have better visualization of the target variable. Prior to the plotting, to have better insight on trend shifting between category, the data is normalized to its mean and standard deviation. The COVID-19 positivity rate is also plotted as important exogenous variable to have better descriptive analysis between variables.

As depicted in Figure 3, there is a unique relationship that can be identified visually by looking into the plotted data between monthly average of positivity rate and sales out data. When the covid-19 positivity rate is increasing, majority the sales out for both skincare and makeup category are decreasing. It explains that covid-19 positivity rate is having negative relationship with product sales, obviously in real world, more people will stay at home during high case of COVID-19 pandemic and do less shopping through outlet store. This line plot is also showing that during the first year of COVID-19, skincare product sales recover faster than makeup product. This is reflecting also during 1st year of COVID-19 less people buying makeup product and shifting to skin care product. But after a year, during 2nd year we can see that makeup category

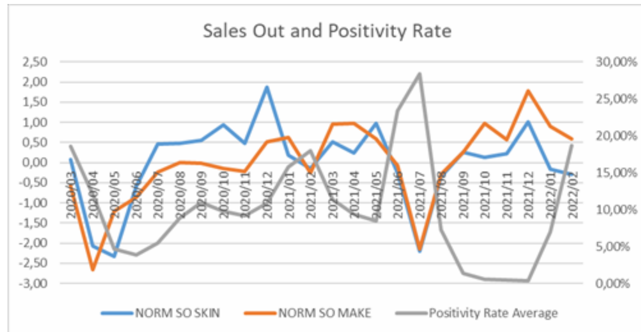
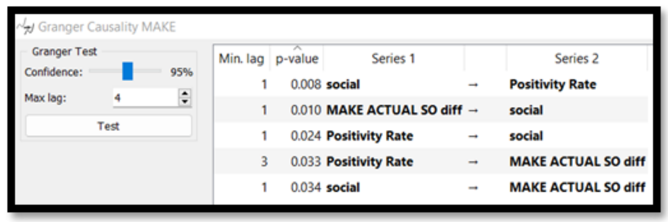


Figure 3: Sales Out and COVID-19 Positivity Rate Data Visualization



Min. lag	p-value	Series 1		Series 2
1	0.008	social	→	Positivity Rate
1	0.010	MAKE ACTUAL SO diff	→	social
1	0.024	Positivity Rate	→	social
3	0.033	Positivity Rate	→	MAKE ACTUAL SO diff
1	0.034	social	→	MAKE ACTUAL SO diff

Figure 4: Granger Causality Test for Makeup Category

Min. lag	p-value	Series 1		Series 2
1	0.006	social	→	Positivity Rate
2	0.029	Positivity Rate	→	social
1	0.043	social	→	SKIN ACTUAL SO diff
4	0.044	Positivity Rate	→	SKIN ACTUAL SO diff

Figure 5: Granger Causality Test for Skincare Category

recover faster than skincare, ultimately with increasing trend since September 2021 onwards. It is clear that as organization, this trend shifting must be addressed with correct and accurate sales and marketing strategy

4 BUILD FORECAST MODEL

4.1 Causality and Stationary Test

One of the forecasting methodologies being used in this research is Vector Autoregression (VAR). Prior to have forecast model by using VAR method, the causality test is being carried out to determine features that will be nominated as key features that will have causality to the target variables. In this research, Granger Causality Test is being performed to determine variables that will be used during forecast modelling using VAR Model.

By looking into Granger Causality test result as showed by Figure 4 and Figure 5 the chosen variables those will be used for build forecast model for skincare are social, positivity rate, and SKIN ACTUAL SO, and for makeup are social, positivity rate, and MAKE

Table 1: Skincare Evaluation Result based on Multiple Forecast Methodologies

SKINCARE	MSE	RMSE	MAPE
Vector Autoregression	3361043	1833,31	38,57%
Random Forest	266434,2	516,17	8,31%
Neural Network	515820,2	654,99	13,74%

Stationary Test SKIN SO DIFF					
Test	Score	P-	C.V.	Stationary?	5,0%
ADF					
No Const	-5,8	0,1000%	-2,1	TRUE	
Const-Only	-5,7	0,2453%	-3,4	TRUE	
Const + Trend	-5,7	0,0000%	-1,6	TRUE	
Const+Trend+Trend^2	-5,5	0,0000%	-1,6	TRUE	
Stationary Test MAKE SO DIFF					
Test	Score	P-	C.V.	Stationary?	5,0%
ADF					
No Const	-6,9	0,1%	-2,1	TRUE	
Const-Only	-6,9	0,1%	-3,4	TRUE	
Const + Trend	-6,9	0,0%	-1,6	TRUE	
Const+Trend+Trend^2	-6,7	0,0%	-1,6	TRUE	

Figure 6: ADF Test for Sales Out Difference Data

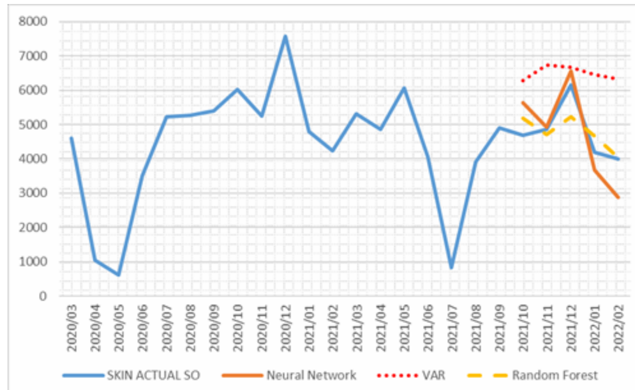


Figure 7: Forecast Result Illustration for Skincare Product Category

ACTUAL SO. After nominating the features, it is also important for variables in which will be included in the modelling to be stationary to ensure removal of significant seasonality and trend. The calculation shows the stationary test by using Augmented Dickey Fuller Test using NumXL Software is returning non-stationarity for sales out data, for these sales out data, the transformation is needed in order to achieve the stationary, hence, we use differencing and run the stationary again and resulted stationary as shown in Figure 6.

4.2 Time Series to Supervised Learning Data Transformation

In machine learning forecast modelling, the time series data needs to be transformed to supervised learning problem, in which will have input variable and target variable. This can be done by using previous observed data as an input. In this case, the sales out from previous three months are being used as input variable (M-1, M-2, and M-3), as well as the sell in value (M-1, M-2, and M-3). Average monthly sales for the past three months are also being used as input variable. In total, there are 26 variables being used in this supervised learning problem, but for each category, there are 16 variables being used for skincare and 16 used for makeup (with 6 sharing variables).

4.3 VAR, Random Forest, and Neural Network

During forecast modelling using Vector Autoregression method, 19 point of time series is being used as training data to build the forecast, and remaining 5 point of time series is being a test data. Later, in the calculation of evaluation forecast model will compare forecast result. Random Forest is being used as one of the methodologies on forecast modelling. The parameter set for random forest is 100 number of trees and since number of variables being used is 15 variables, the number of attributes being considered in each split is $\sqrt{15}$ which is approximately 4 attributes in each split. Neural Network is also being used as one of supervised learning method in this research. After several simulation run using Orange Data Mining Software, the parameter used in neural network is only one layer used with 1000 neuron and ReLu activation function.

5 RESULT AND DISCUSSIONS

The data visualization for above forecast method can be seen in Figure 7 and Figure 8. The best method can be explained visually by this graph. The graph display vary result between model but all model looks can explain the inclination on near to end of 2021 and declining sales in the beginning of 2022.

Evaluation of the model is carried out to determine best model for each product category by using certain error measurement and visualization. Table 1 and Table 2 shows the error calculation in this research. Error parameters being used in this research are common KPIs for determining forecast performance, which are Mean Square Error (MSE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE).

For skincare, machine learning provides more promising result compared to time series forecast with regards to performance. The random forest returned lowest RMSE and lowest MAPE, in this case giving us better forecast accuracy for 5 months observation. In the other hand, the neural network returned moderate number of error measurement but from the visualization we can see that the neural network can explain the seasonality very well since it is having closest line pattern compared to the actual sales.

Table 2: Makeup Evaluation Result based on Multiple Forecast Methodologies

MAKEUP	MSE	RMSE	MAPE
Vector Autoregression	10570	102,81	8,67%
Random Forest	59891	244,73	32,91%
Neural Network	59780	270,03	19,08%

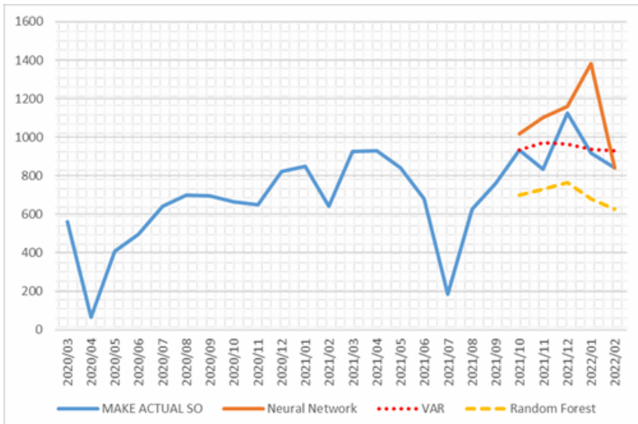


Figure 8: Forecast Result Illustration for Makeup Product Category

For makeup, the time series forecast perform better compared to machine learning based on error measurement. VAR returned lowest number of RMSE and MAPE. From the visualization, VAR-based forecast model looks flat and not really explaining the seasonality slope well in the other hand random forest and neural network are also not showing satisfactory result and not explaining the seasonality well.

6 CONCLUSIONS

From the forecast model generated previously, data visualization is considerably important. From the result of error measurement and data visualization provided in previous section, both analysis of error rate and data visualization are equally important than single analysis. The smallest error rate doesn't imply on how close the seasonality and pattern plotted in the visualization. As an example, on skincare forecast result, if only considering error measurement, random forest might be nominated as best method, but when evaluation is being done through visualization on plotted data, the neural network performs better than random forest by reflecting better and closer trend line compare to the actual numbers.

The data visualization may give important information to the expert user on decision making process other than error measurement. In order to provide better data-information-knowledge to the business user as the result of this research, then below parameter are being considered as part of data visualization that may contribute to expert user knowledge creation for better sales and marketing strategy:

- Plot of actual historical sales out actual data (year to date).
- Plot of sales out forecast data (YTD) and 12 months forecast period (future).
- Error measurement (MAPE) & Forecast Accuracy (100%-MAPE).
- Exogenous variable – COVID-19 positivity rate, Social Restriction Level.

This research contributes to the organization by providing first ever multivariate forecasting methodology to the business planner by including COVID-19 data into forecast model creation as well as providing normalized data visualization across category. The provided data visualization may complement current sales forecast quarter review as additional knowledge in the consensus meeting. By providing additional information on how external data (known as exogenous variable) as well as modelled tacit knowledge may give new perspective on sales forecast determination. In the other hand, for machine learning based forecast model, since COVID-19 positivity rate considered as input (independent variable), expert user might be able to adjust COVID-19 data as self-input data to investigate the sales out forecast number. This will provide additional information to the expert user that would be beneficial for rational decision making on top of intuition-based decision making in which common in forecast generation.

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